

Revised Mean-Field Search Theory

Empirical peer propagation versus finite epistemic persistence

29 August 2026

Purpose

The earlier screening report used a classical x^q social-contagion term. That is too strong an assumption for the LLM game. In the actual runtime, q is simply the number of social peer slots visible to the focal agent; the LLM is not required to obey a classical q -voter unanimity rule. This revision removes that assumption. The peer response is represented by a quantity that can be measured directly from the stored LLM transitions.

The goal is deliberately narrow: determine whether a controller-created false majority can continue to propagate through ordinary peers after the controller is removed, while true proofs continue to spread but are not perfectly persistent.

1 Definitions: all variables before the model

We use only the following quantities.

Symbol	Meaning
x_t	Fraction of the population voting for the false target at round t .
ϕ_t	Fraction of agents for whom the complete true proof is currently available at round t .
q	Number of ordinary social peer slots shown to a focal agent when the controller is absent. In the current experiments, $q = 2$ or $q = 3$.
ρ	Probability that one true supporting fact remains available from one round to the next. $\rho = 1$ is the current permanent-information game.
L	Number of true supporting facts needed for a complete proof. We propose fixing $L = 3$ rather than sweeping it.
a	Per-round probability that an agent without a complete active proof acquires one through information propagation. This is estimated from existing trajectories, not swept.
$g_q(x)$	Empirical probability that an agent without the complete proof votes false after a peer-only update when the population false share is x . This is measured from LLM transitions.
A_q	Maximum majority-level peer amplification, defined below. It summarizes how strongly ordinary agents can reproduce a false majority without the controller.

The controller does not appear in the post-release equations. Its role is to create the initial false seed x_0 . Study 09b has already shown that the controller can push the population into the approximate range $x \approx 0.58$ – 0.67 in some episodes. The present question is what happens after that intervention is removed.

2 1. Epistemic persistence: the truth process

If each of the L required facts persists independently with probability ρ , then a complete proof survives one round with probability

$$P = \rho^L. \quad (1)$$

A minimal round-level mean-field equation for complete-proof availability is

$$\boxed{\phi_{t+1} = P\phi_t + a(1 - \phi_t).} \quad (2)$$

The first term is retained proof; the second is new proof acquisition among agents that do not currently have the complete proof.

Equation (2) has the exact stationary solution

$$\boxed{\phi^* = \frac{a}{1 - P + a} = \frac{a}{1 - \rho^L + a}.} \quad (3)$$

Therefore

$$\boxed{1 - \phi^* = \frac{1 - \rho^L}{1 - \rho^L + a}} \quad (4)$$

is the stationary fraction of agents that remain epistemically susceptible because they do not currently have the complete proof.

The permanent-information limit is immediate:

$$\rho = 1 \implies \phi^* = 1.$$

This reproduces the structural behavior seen in Study 09b: if true information is permanent, complete proof eventually becomes absorbing.

Using the observed rough timing $\phi \approx 0 \rightarrow 0.75$ in about five rounds in the no-loss regime gives

$$0.75 \approx 1 - (1 - a)^5 \implies \boxed{a \approx 0.242 \text{ per round.}} \quad (5)$$

This is only a screening calibration; the final value should be re-estimated directly from the stored round transitions.

3 2. Social propagation: measure the LLM response instead of assuming x^q

For a peer-only update, suppose the focal agent sees q ordinary peers. If k of those peers currently vote false, define

$$p_{q,k} = \Pr(\text{focal votes false next} \mid k \text{ false peers out of } q, \text{ no complete proof}). \quad (6)$$

These probabilities can be estimated directly from stored micro-slot transitions or from a very small peer-only calibration. They are the LLM's actual social response.

Under well-mixed peer sampling, the probability that an incomplete-proof focal agent votes false when the population false share is x is

$$\boxed{g_q(x) = \sum_{k=0}^q \binom{q}{k} x^k (1 - x)^{q-k} p_{q,k}.} \quad (7)$$

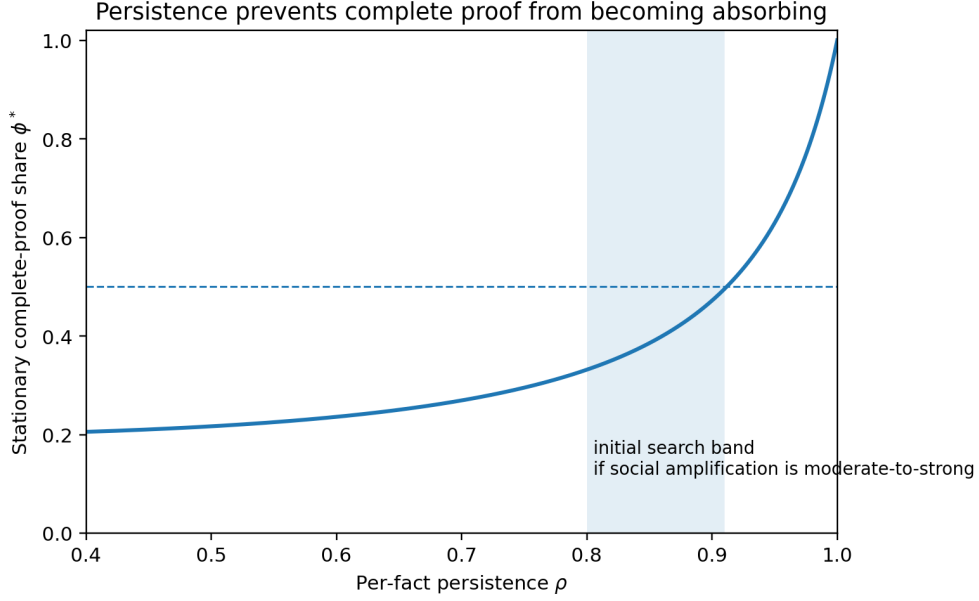


Figure 1: Exact stationary complete-proof fraction from Eq. (3), using $L = 3$ and $a = 0.242$. At $\rho = 1$, complete proof becomes absorbing. Lower persistence maintains a nonzero susceptible population.

This equation is important because q now has exactly its runtime meaning: it is the number of peer slots. No unanimity rule is assumed. If the LLM reacts weakly to peer agreement, the measured $p_{q,k}$ will show that. If it strongly amplifies a majority, the same coefficients will show that.

After the controller is removed, a minimal false-share map is

$$x_{t+1} \approx (1 - \phi_t)g_q(x_t). \quad (8)$$

For screening, fully informed agents are treated as truth-directed, consistent with the near-perfect full-information calibration. A small residual full-proof error can be added later if needed.

4 3. The key result: when can a false majority sustain itself?

We care specifically about a *macroscopic false majority*, so restrict attention to $x \geq 1/2$. Define

$$A_q = \max_{x \in [1/2, 1]} \frac{g_q(x)}{x}. \quad (9)$$

A_q has a simple interpretation:

A_q is the strongest one-round amplification that ordinary peer interaction can give an already macroscopic false state.

If $A_q = 1$, peers do not amplify false support. If $A_q > 1$, there is some majority-level state for which peer interaction increases false support among agents without the complete proof.

At stationarity, Eq. (8) can amplify a false majority only if

$$A_q(1 - \phi^*) > 1. \quad (10)$$

Substituting Eq. (4) gives the explicit persistence condition

$$A_q \frac{1 - \rho^L}{1 - \rho^L + a} > 1. \quad (11)$$

Solving for the critical persistence yields

$$\rho_c = \left(1 - \frac{a}{A_q - 1}\right)^{1/L}, \quad (12)$$

provided $A_q > 1 + a$. If

$$A_q \leq 1 + a, \quad (13)$$

then even the strongest possible loss of proof in this minimal model does not produce a peer-sustained false majority. In that case persistence is not the missing mechanism; the ordinary social response itself is too weak.

This is the main decision rule of the revised theory.

5 4. Numerical location of the search region

Using $L = 3$ and the rough existing estimate $a = 0.242$, the threshold becomes a direct function of one quantity that we must measure: A_q .

Measured A_q	Predicted ρ_c	Interpretation
1.30	0.578	Weak social amplification; strong proof turnover required
1.40	0.734	
1.50	0.802	
1.60	0.842	Moderate amplification
1.75	0.878	Strong amplification
2.00	0.912	Absolute majority-level upper bound

The last row is a useful hard bound. Because $g_q(x) \leq 1$ and we restrict to $x \geq 1/2$, necessarily $A_q \leq 2$. With the current calibration,

$$\rho \gtrsim 0.912 \implies \text{no peer-sustained false majority in this closure, even under maximal social amplification.} \quad (14)$$

Thus the earlier $\rho \approx 0.95$ screening region was too optimistic; it was a consequence of imposing a classical x^q law rather than using the actual LLM peer response.

6 5. Exactly what should be measured before another LLM sweep?

Do not sweep persistence yet. First estimate the few peer-response numbers needed for g_q .

For the preferred first study, fix $q = 2$. Then only three response probabilities are required:

$$p_{2,0}, \quad p_{2,1}, \quad p_{2,2}.$$

They answer: when an incomplete-proof focal agent sees zero, one, or two false peers, how often does it vote false next?

From them,

$$g_2(x) = (1 - x)^2 p_{2,0} + 2x(1 - x)p_{2,1} + x^2 p_{2,2},$$

and A_2 follows directly from Eq. (9). No new theory parameter is introduced. A_2 is measured once from the LLM system.

The existing Study-09b micro-slot data may already be sufficient for a first estimate if enough NO_OP/peer-only updates exist. If not, a tiny no-controller calibration is enough. The same procedure for $q = 3$ would require four coefficients, but $q = 3$ should be a validation condition rather than another main axis.

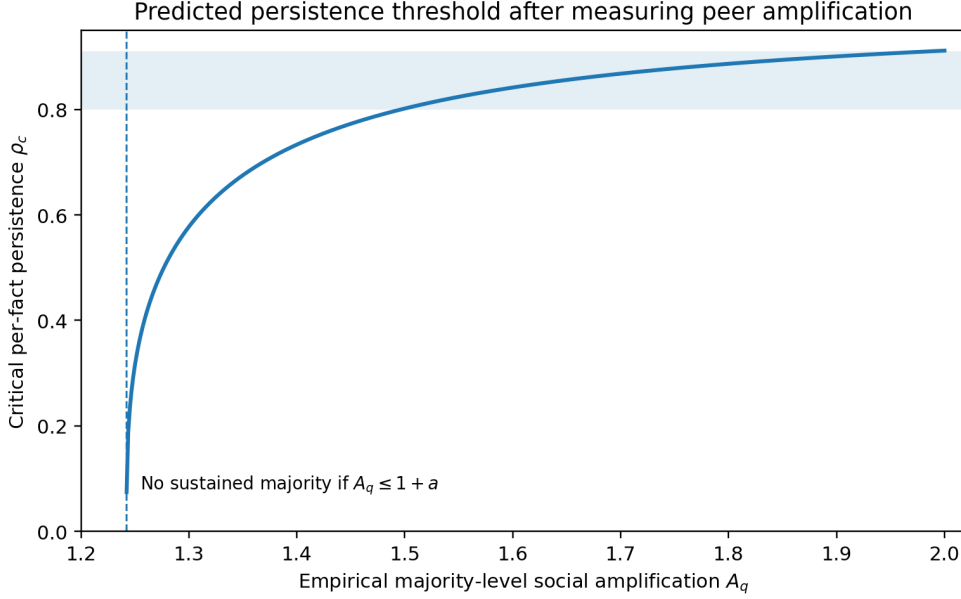


Figure 2: Closed-form critical persistence as a function of the empirically measured majority-level social amplification A_q . The shaded band $\rho \approx 0.80\text{--}0.91$ is the most relevant region if the LLM exhibits moderate-to-strong peer amplification, $A_q \approx 1.5\text{--}2$.

7 6. The minimal experimental search

To avoid parameter explosion, the first persistence experiment should fix everything except persistence:

$$q = 2, \quad L = 3, \quad N = 12,$$

with the same controller sensing and soft-policy settings already used in the strong Study-09b condition. Use $b = N$ only to create the seed, then release the controller.

The workflow is:

1. Estimate a and A_2 from stored data.
2. Compute ρ_c from Eq. (12).
3. Run a seed-and-release bracket at approximately

$$\rho \in \{1, \rho_c + 0.03, \rho_c, \rho_c - 0.03\}$$

(clipped to $[0, 1]$).

4. The controller acts only long enough to create a false seed near the already observed $x \approx 0.6$ range, then is forced to NO_OP.
5. The decisive endpoint is whether x_t remains above $1/2$ or grows through peer-only dynamics while ϕ_t remains nonzero and true facts continue to circulate.

If the empirical A_2 falls roughly in the moderate-to-strong range $1.5\text{--}2$, the calculations place the first useful persistence bracket around

$$\rho \approx 0.80\text{--}0.91. \tag{15}$$

If A_2 is lower, the formula automatically moves the search to lower persistence. If $A_2 \leq 1.242$, persistence alone is not predicted to create the desired self-propagating false-majority regime at all.

Only after this regime is found should the control budget b/N be varied to locate the minimum seed-generating control expenditure. That later step is where control efficiency becomes the central observable.

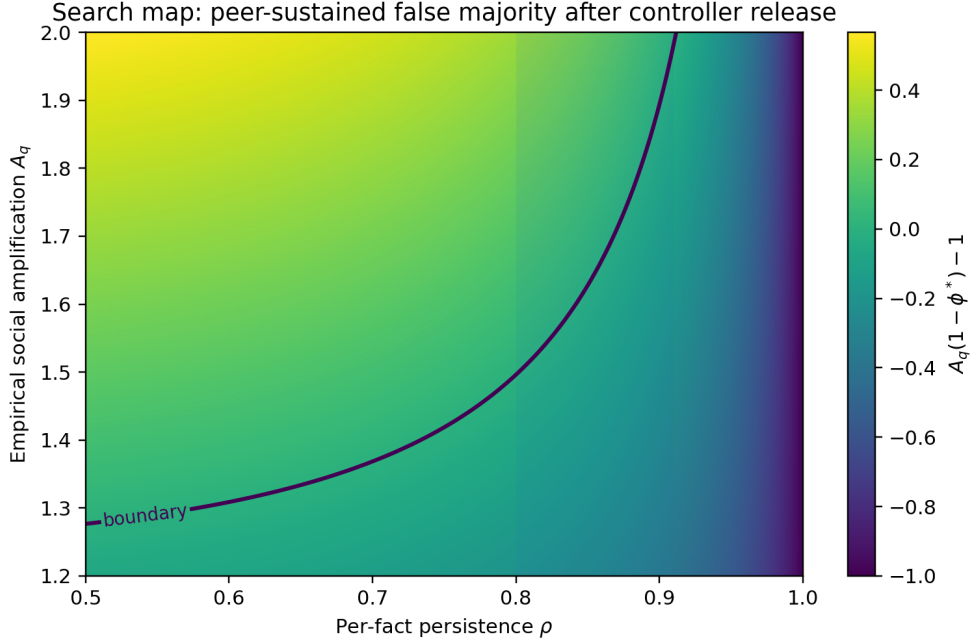


Figure 3: Analytic phase map from Eq. (10). Above/below the boundary the same controller-created seed is predicted to be socially self-sustaining/decaying after controller release, subject to the mean-field assumptions.

8 Main results in two paragraphs

The revised theory has one central result. Let A_q measure how strongly ordinary peers amplify an already macroscopic false state, and let ρ measure how persistently each true fact remains available. Complete proof no longer becomes absorbing when $\rho < 1$; instead, the stationary fully informed fraction is $\phi^* = a/(1 - \rho^L + a)$. A controller-created false majority can then sustain itself after the controller is removed only if peer amplification overcomes the remaining fully informed population: $A_q(1 - \phi^*) > 1$. This gives the closed-form threshold $\rho_c = (1 - a/(A_q - 1))^{1/L}$. Importantly, A_q is measured from the LLM peer transitions; it is not another parameter to sweep, and the theory no longer assumes a classical x^q voting law.

For the current $L = 3$ setup and the rough Study-09b proof-acquisition estimate $a \approx 0.24$, the first useful region is likely $\rho \approx 0.80\text{--}0.91$ if peer amplification is moderate to strong ($A_q \approx 1.5\text{--}2$). Moreover, $\rho \gtrsim 0.912$ cannot sustain a false majority in this minimal closure even under the strongest possible majority-level peer amplification. Therefore the next decision should be: estimate A_2 from existing peer-only transitions, calculate ρ_c , and run only a few $q = 2$ seed-and-release simulations immediately around that predicted boundary. If the false majority survives after controller release, then the agents - not continued external control - are propagating the false collective state faster than epistemic repair can remove it.